

Data Mining

Decision tree classifier





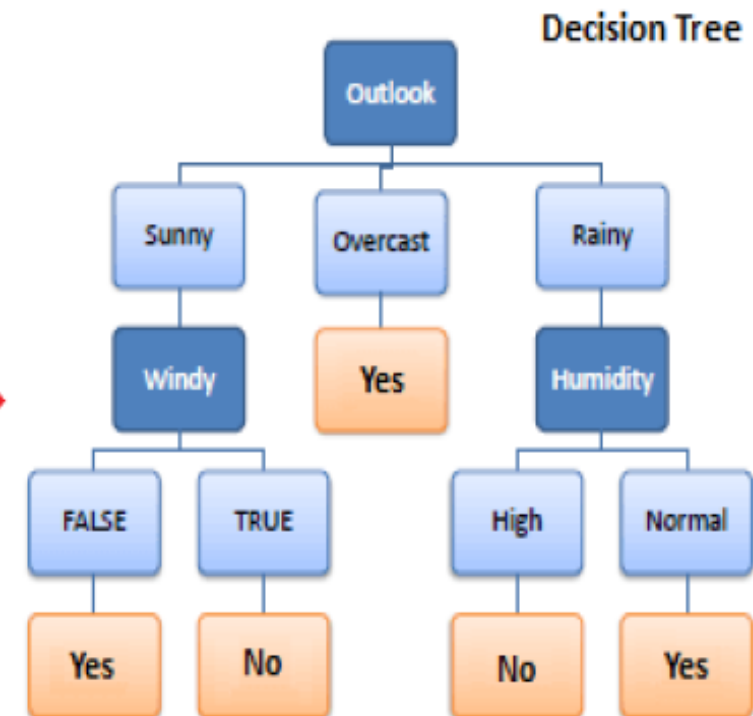
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Decision Tree Classifier Example

Predictors				Target
Outlook	Temp.	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No





Decision Tree Classifier (cont.)

Decision Tree Consists of:

- **Nodes:**

- *Test for the value of a certain attribute.*

- **Edges:**

- *Correspond to the outcome of a test.*
- *Connect to the next node or leaf.*

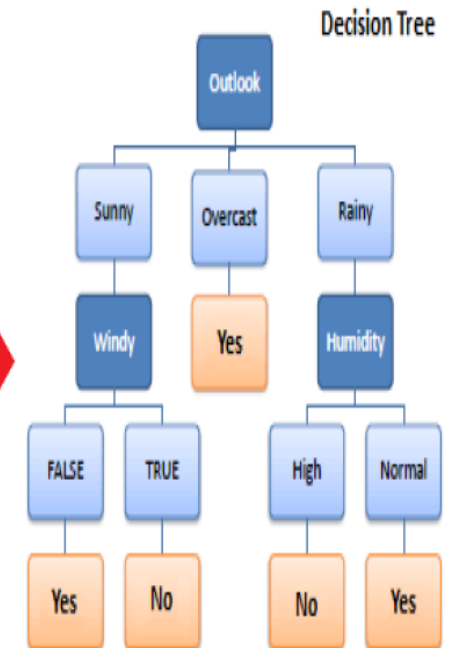
- **Leaves:**

- *Terminal node that predict the outcome.*

Decision Tree Algorithm

- Decision tree Algorithm transform data into a tree and the prediction of new instance will be done using tree.
- Let's see set of equations needed to apply decision tree on data and get tree.

Predictors				Target
Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No





Decision Tree Classifier Algorithm (cont.)

First Equation:

– Information Gain(IG)

- *The information gain measure the relation between each feature and the goal.*
- *we agree that decision tree is a classification algorithm, so it apply on data that has a class label like in previous table.*
- *So first we measure the relation of each feature with the goal using **information Gain(IG)** and the feature which has the **highest** information gain will be the **root** of tree.*



Decision Tree Classifier Algorithm (cont.)

First Equation:

- Information Gain(IG)

- *Information Gain Equation:*

$$Gain(T, X) = Entropy(T) - Entropy(T, X)$$

Where => *T* means Target/Class Label.

X means Features.

- From previous table we have four features so we will calculate IG Four times and select feature with the **highest** IG to be **root** of tree.



Decision Tree Classifier Algorithm (cont.)

Second Equation:

- From IG equation we can see that in order to calculate IG we need to calculate entropy at first.

$$Gain(T, X) = Entropy(T) - Entropy(T, X)$$

- Note that:
 - Entropy measure “how much information included” so we need it to tell us the feature which has more information.



Decision Tree Classifier Algorithm (cont.)

Second Equation:

$$Gain(T, X) = Entropy(T) - Entropy(T, X)$$

- *To Calculate entropy for one element $Entropy(T)$:*

$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

- *To Calculate Entropy for Two elements $Entropy(T, X)$:*

$$E(T, X) = \sum_{c \in X} P(c) E(c)$$



Decision Tree Classifier Algorithm Equations

So, The Equations need for Decision tree are:

1 $Gain(T, X) = Entropy(T) - Entropy(T, X)$

2
$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

3
$$E(T, X) = \sum_{c \in X} P(c) E(c)$$



Decision Tree Algorithm Steps.

➤ Steps for Algorithm:

- Calculate the entropy of the target
- calculate the information gain of the pair (attribute, target)
- Select the attribute with the highest gain to be the node
- attributes with zero are leaves.
- More splitting

Given The Following Table and apply Decision Tree

Predictors				Target
Outlook	Temp.	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

Algorithm Steps:

– Step 1:

Calculate entropy for target (Play golf) as follow:

$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

Play Golf	
Yes	No
9	5

Entropy(PlayGolf) = Entropy (5,9)
= Entropy (0.36, 0.64)
= - (0.36 log₂ 0.36) - (0.64 log₂ 0.64)
= 0.94

Algorithm Steps (cont.)

– Step 2:

Calculate entropy for target and each feature as follow:

Note that

$$E(3,2) \Rightarrow -[3/5 \log(3/5) + 2/5 \log(2/5)]$$

$$E(T, X) = \sum_{c \in X} P(c) E(c)$$

		Play Golf		
		Yes	No	
Outlook	Sunny	3	2	5
	Overcast	4	0	4
	Rainy	2	3	5
				14

$$\begin{aligned} E(\text{PlayGolf}, \text{Outlook}) &= P(\text{Sunny}) * E(3,2) + P(\text{Overcast}) * E(4,0) + P(\text{Rainy}) * E(2,3) \\ &= (5/14) * 0.971 + (4/14) * 0.0 + (5/14) * 0.971 \\ &= 0.693 \end{aligned}$$



Algorithm Steps (cont.)

- Step 3:

- Calculate Information Gain(IG):

Subtract Entropy(T) from Entropy(T,X) to get information gain(IG) as Follow:

$$\begin{aligned} G(\text{PlayGolf, Outlook}) &= E(\text{PlayGolf}) - E(\text{PlayGolf, Outlook}) \\ &= 0.940 - 0.693 = 0.247 \end{aligned}$$

Algorithm Steps (cont.)

- **Step 4:**
 - Repeat step 2,3 for all other features to get IG as follow:
 - Calculate E(T, Every feature) then subtract from E(T) to get information Gain (IG)

		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3
		Gain = 0.247	

		Play Golf	
		Yes	No
Temp.	Hot	2	2
	Mild	4	2
	Cool	3	1
		Gain = 0.029	

		Play Golf	
		Yes	No
Humidity	High	3	4
	Normal	6	1
		Gain = 0.152	

		Play Golf	
		Yes	No
Windy	False	6	2
	True	3	3
		Gain = 0.048	

$$Gain(T, X) = Entropy(T) - Entropy(T, X)$$

Algorithm Steps (cont.)

- **Step 5:**
 - Select feature with **highest** information gain to be **root** of tree.
 - In this case **outlook** has **highest** information gain so it will be the **root**.

		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3
Gain = 0.247			

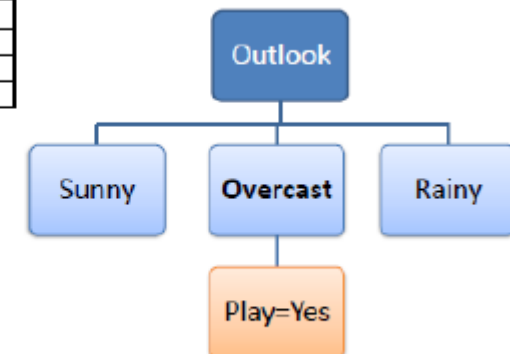
Outlook				
Sunny				
Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Mild	High	FALSE	Yes
Sunny	Cool	Normal	FALSE	Yes
Sunny	Cool	Normal	TRUE	No
Sunny	Mild	Normal	FALSE	Yes
Sunny	Mild	High	TRUE	No
Overcast				
Overcast	Hot	High	FALSE	Yes
Overcast	Cool	Normal	TRUE	Yes
Overcast	Mild	High	TRUE	Yes
Overcast	Hot	Normal	FALSE	Yes
Rainy				
Rainy	Hot	High	FALSE	No
Rainy	Hot	High	TRUE	No
Rainy	Mild	High	FALSE	No
Rainy	Cool	Normal	FALSE	Yes
Rainy	Mild	Normal	TRUE	Yes

Algorithm Steps (cont.)

- **Step 6:**
 - Divide the table into 3 small tables based on **variation** under **outlook** feature.
 - We will see that in **overcast** table all instances classify as **yes** and Entropy is **0**, so this branch is finish and labeled as **yes**

Outlook	Temp	Humidity	Windy	Play Golf
Sunny	Mild	High	FALSE	Yes
Sunny	Cool	Normal	FALSE	Yes
Sunny	Cool	Normal	TRUE	No
Sunny	Mild	Normal	FALSE	Yes
Sunny	Mild	High	TRUE	No
Overcast	Hot	High	FALSE	Yes
Overcast	Cool	Normal	TRUE	Yes
Overcast	Mild	High	TRUE	Yes
Overcast	Hot	Normal	FALSE	Yes
Rainy	Hot	High	FALSE	No
Rainy	Hot	High	TRUE	No
Rainy	Mild	High	FALSE	No
Rainy	Cool	Normal	FALSE	Yes
Rainy	Mild	Normal	TRUE	Yes

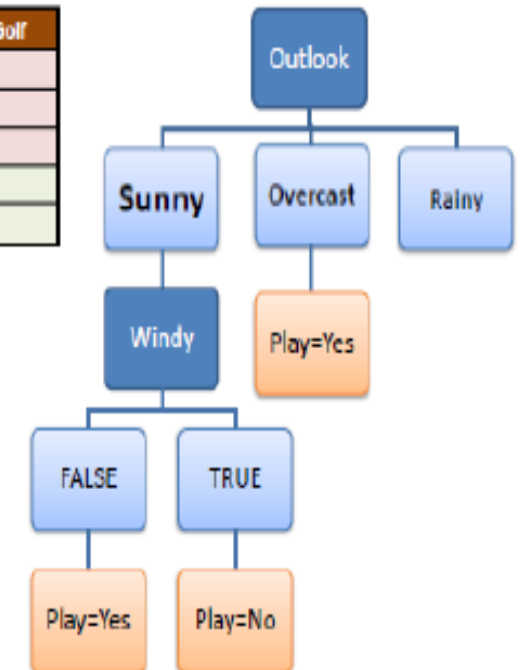
Temp	Humidity	Windy	Play Golf
Hot	High	FALSE	Yes
Cool	Normal	TRUE	Yes
Mild	High	TRUE	Yes
Hot	Normal	FALSE	Yes



Algorithm Steps (cont.)

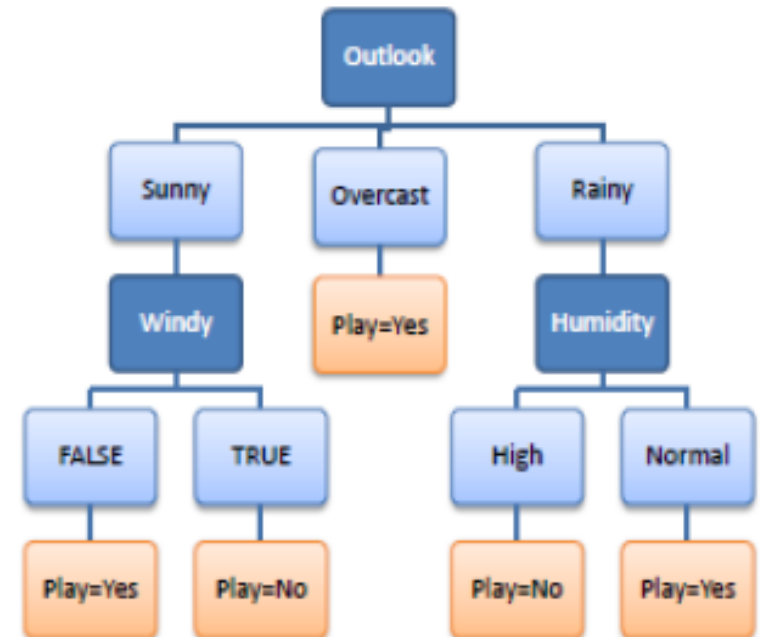
- **Step 7:**
 - After classify **overcast** as **yes**, we still have two branches (sunny, rainy) that need **further** work.
 - Treat each table of remain 2 tables as a **separate** table and **repeat** steps from step 1 calculate **E(target)** of new table then calculate **E(T, Every feature)** then calculate **IG**.
 - Select **feature** with **highest** IG to be the next node of tree.

Temp	Humidity	Windy	Play Golf
Mild	High	FALSE	Yes
Cool	Normal	FALSE	Yes
Mild	Normal	FALSE	Yes
Cool	Normal	TRUE	No
Mild	High	TRUE	No



Algorithm Steps (cont.)

- **Step 8:**
 - Continue till the tree is complete as shown in figure:
 - We can observe that **temp feature** does not **appear** in tree because it's an **irrelevant feature** and it should be **removed** before starting the algorithm using any dimension reduction technique like **PCA**.
 - Therefore, we say that more instance should be as a **row** not as a **column**.



Algorithm Steps (cont.)

- **Step 9:**

- Construct decision rule from decision tree.
- We have 5 leaves so we will have 5 rules trace tree from root and type all rules till we reach each leaf.

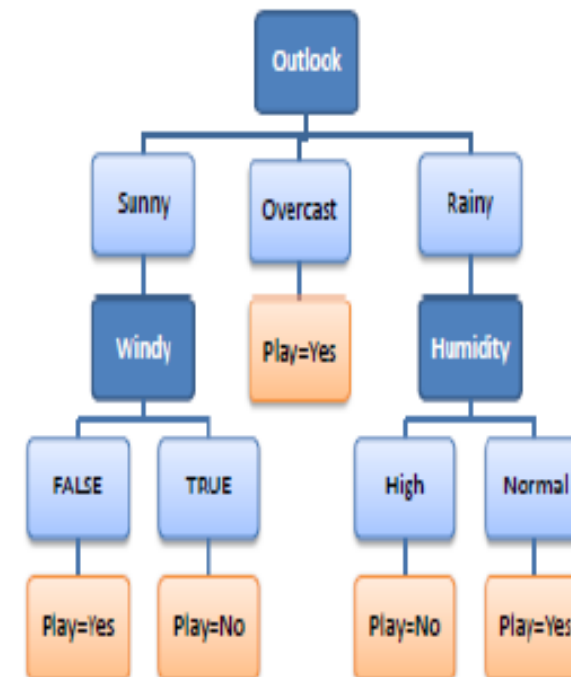
R_1 : IF (Outlook=Sunny) AND (Windy=FALSE) THEN Play=Yes

R_2 : IF (Outlook=Sunny) AND (Windy=TRUE) THEN Play=No

R_3 : IF (Outlook=Overcast) THEN Play=Yes

R_4 : IF (Outlook=Rainy) AND (Humidity=High) THEN Play=No

R_5 : IF (Outlook=Rain) AND (Humidity=Normal) THEN Play=Yes





Any Questions?





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THANK

YOU!

